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RESEARCH ARTICLE

Dual-Stage Adaptive Fuzzy MPC for Autonomous Vehicle Obstacle Avoidance

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ABSTRACT Robust trajectory tracking of autonomous vehicles under varying speeds and uncertain road adhesion remains challenging. This paper proposes a dual-stage adaptive fuzzy model predictive control (MPC) framework that tunes the MPC cost weights online based on vehicle states and road friction conditions. A fuzzy inference system (FIS) is optimized offline using particle swarm optimization (PSO) followed by Bayesian optimization (BO), and then deployed for real-time weight adaptation. Extensive simulations in obstacle-rich scenarios with diverse speeds and friction coefficients evaluate the proposed controller against baseline MPC, proportional–integral–derivative (PID) control, linear quadratic regulator (LQR), and nonlinear MPC (NMPC). Compared with baseline MPC, the proposed method reduces lateral tracking error by 6%–38% (scenario-dependent), improves heading accuracy by up to 49%, and decreases steering effort by 12%–39%. It further achieves lower tracking errors and smoother steering than PID, while providing a competitive accuracy–effort trade-off relative to LQR and NMPC in obstacle-rich conditions. The results demonstrate improved tracking accuracy, control smoothness, and robustness for autonomous vehicles operating in complex environments.

INDEX TERMS Adaptive fuzzy MPC, autonomous vehicles, PSO, Bayesian.

I. INTRODUCTION

As vehicle ownership grows and traffic environments become increasingly complex, traditional driving modes face mounting challenges, including congestion, environmental pollution, and safety risks [1]. Autonomous driving technology, as an interdisciplinary product of artificial intelligence [2], [3], sensor fusion [4], communication, and advanced control, has been recognized as a promising approach to address these issues [5]. Autonomous vehicles (AVs) are expected to significantly improve the efficiency and safety of road transportation, mitigate human-induced accidents, enhance user experience, and provide mobility solutions for people with limited physical abilities. Among the core technologies enabling autonomous driving, path planning and trajectory tracking control are fundamental for safe and efficient vehicle operation [6]. This is especially true in complex and dynamic

environments, where timely and precise obstacle avoidance is critical for system reliability and safety.

Model Predictive Control (MPC) has emerged as a mainstream approach for trajectory tracking and obstacle avoidance in AVs [7], owing to its capability to handle multi-variable constraints and optimize control actions over a prediction horizon. The demand for high reliability and robustness in autonomous systems mirrors challenges in broader complex control engineering. For instance, advanced theoretical frameworks, such as the synchronization of fractional reaction-diffusion complex networks [8] and the fault-tolerant synchronization of switched complex networks [9], have been extensively developed to guarantee system stability under severe disturbances and faults. Inspired by these advancements, researchers have continuously refined the MPC framework to address the nonlinearities, uncertainties, and real-time demands inherent in complex driving scenarios. Notably, He et al. [10] introduced a neural network-based event-triggered MPC strategy that

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utilizes event-driven re-planning and neural approximations for vehicle model uncertainties, thereby reducing triggering frequency and computational burden while enhancing obstacle avoidance efficiency. Wu et al. [11] proposed an event-triggered dual-layer MPC strategy that combines an upper-level safety planner with a lower-level preview-based MPC controller. This event-triggered mechanism and time-varying prediction horizon result in improved tracking accuracy and real-time performance compared to standard MPC. Zheng et al. [12] integrated a fuzzy MPC with soft constraints on tire slip angle and a three-dimensional risk potential field, enabling adaptive horizon tuning and yielding smooth, accurate trajectories even at high speeds. Further, Sun et al. [13] developed a system that couples MPC with an extended Kalman filter for trajectory tracking and combines MPC with artificial potential field methods in the re-planning module, resulting in enhanced obstacle avoidance and tracking precision. Adaptive and data-driven paradigms are also advancing. For example, Kang et al. [14] proposed an adaptive trajectory-tracking controller with grey relational analysis and Fourier approximation for offline optimization. Additionally, Nasri et al. [15] introduced a framework employing kernel density estimation and Kernel Principal Component Analysis (KPCA) for uncertainty quantification, delivering substantial improvements in tracking errors and real-time feasibility.

Recent studies have further expanded the applicability of MPC in obstacle avoidance scenarios, with a strong emerging focus on adaptive mechanisms and cooperative control to handle complex and extreme dynamics. For instance, to enhance lateral tracking robustness under varying operating conditions, a recent study proposed a fuzzy adaptive dynamic MPC strategy that utilizes a whale optimization algorithm for adaptive horizon scheduling alongside a fuzzy regulator to dynamically adjust objective function weights [16]. Similarly, to address extreme maneuvers such as emergency lane changes, another advanced cooperative framework integrates an MPC trajectory tracker with sliding mode-based active front-wheel steering and direct yaw moment control to maximize vehicle stability [17]. Alongside these developments, Gratzner et al. [18] developed a dual-layer MPC framework combining a fast quadratic MPC for local avoidance and a mixed-integer MPC for global trajectory updates, demonstrating high efficiency and collision-free navigation through Vehicle-to-Everything (V2X)-enhanced predictions. Ma et al. [19] proposed a hybrid control scheme that integrates nonlinear MPC for dynamic obstacle avoidance with linear time-varying MPC for trajectory tracking, improving driving stability and comfort. Zhang et al. [20] introduced a two-layer MPC approach that adaptively adjusts obstacle avoidance weights based on road and speed conditions, effectively reducing over-avoidance and tracking deviation. Reiter et al. [21] reformulated the MPC problem using both Cartesian and Frenet frames, enabling more accurate obstacle representation and faster optimization for nonlinear avoidance tasks.

Despite these advances, several challenges remain. The trade-off between model fidelity and computational feasibility persists: complex models hinder real-time control, whereas oversimplified models compromise control accuracy, particularly under extreme or uncertain conditions [15]. Fixed cost function weights often restrict adaptability across diverse driving environments. Although adaptive MPC schemes and fuzzy logic control have shown promise for dynamic weight tuning and handling uncertainties, the efficient, global optimization of fuzzy controller parameters—especially membership functions—remains a bottleneck in high-dimensional, multi-objective tasks. Furthermore, the real-time coordination of multiple, sometimes conflicting, objectives—such as safety, comfort, tracking accuracy, and vehicle stability—continues to present substantial technical hurdles.

To address these issues, this paper proposes an Adaptive Fuzzy Model Predictive Control (AFMPC) algorithm for autonomous vehicle obstacle avoidance. The core innovation of this approach is the synergy between offline dual-stage optimization of a fuzzy inference system (FIS) and online fuzzy adaptation of MPC cost-function weights. Specifically, in the offline stage, particle swarm optimization (PSO) is employed for global coarse tuning, followed by Bayesian optimization (BO) to fine-tune the FIS membership functions. This dual-stage approach yields a well-calibrated inference mechanism under representative operating conditions. During online operation, the optimized FIS dynamically adjusts MPC cost weights in response to real-time road adhesion, vehicle states, and obstacle-related cues, thereby balancing tracking accuracy, stability, and control smoothness across diverse scenarios. The AFMPC framework is validated in simulation across multiple obstacle-rich scenarios. Its performance is benchmarked against a fixed-weight baseline MPC, a Proportional–Integral–Derivative (PID) controller, a Linear Quadratic Regulator (LQR) as a classical linear reference, and a Nonlinear Model Predictive Control (NMPC) algorithm as a high-fidelity reference. The results demonstrate consistent improvements in trajectory-tracking accuracy, vehicle stability, and steering smoothness across varying speeds and friction levels.

The main contributions of this work are as follows:

- Proposing a dual-stage offline tuning framework (PSO $\rightarrow$ BO) for optimizing FIS membership functions, enabling reliable and well-conditioned fuzzy inference for subsequent online adaptation.
- Developing a multi-source, real-time fuzzy adaptation mechanism that adjusts MPC cost weights according to road adhesion, vehicle states, and obstacle-induced demands, achieving a robust accuracy–smoothness trade-off.
- Validating the proposed AFMPC through comprehensive simulations across five representative scenarios, benchmarking against baseline MPC, PID, LQR, and NMPC, and demonstrating improved tracking

performance and control smoothness under obstacle-rich driving conditions.

The rest of this paper is organized as follows: Section II describes the vehicle dynamics modeling. Section III presents the design of the AFMPC controller, including the fuzzy adaptive weight adjustment mechanism. Section IV details the two-stage optimization of FIS membership functions. Section V provides simulation results and comparative analyses. Section 6 concludes the paper and discusses future research directions.

II. PROBLEM FORMULATION AND SYSTEM MODELING

This section formulates the obstacle-avoidance problem and establishes the system models required by the proposed dual-stage adaptive fuzzy MPC controller. Specifically, we present the vehicle dynamics, the environment representation, and the control objectives, which together constitute the mathematical foundation of the predictive control strategy. The vehicle model is introduced first.

A. KINEMATIC AND DYNAMIC MODELING OF AUTONOMOUS VEHICLES

Accurate mathematical modeling of vehicle motion is essential for autonomous driving control [22]. To develop a robust and real-time applicable obstacle avoidance framework, this study employs three distinct vehicle models of varying fidelity, each strictly tailored to a specific functional requirement within the proposed control architecture. First, the kinematic model is utilized to establish the basic geometric relationships and generate reference trajectories, as it provides a computationally efficient representation of the vehicle motion at low speeds or small steering angles without the complexities of tire force coupling. Second, a linearized 2-DOF dynamic model is adopted as the internal prediction model for the online Adaptive MPC controller. By linearizing the complex vehicle dynamics, the predictive control sequence can be formulated as a convex Quadratic Program (QP), thereby guaranteeing the real-time computational efficiency and strict execution deadlines required for onboard autonomous driving systems. Finally, a high-fidelity nonlinear dynamic model, which incorporates the Brush tire model to accurately capture lateral tire force saturation and highly coupled nonlinearities under large slip conditions, is employed for two critical purposes: it serves as the rigorous simulation plant environment (replacing the real vehicle) for closed-loop validation, and it acts as the high-precision evaluation baseline for the offline PSO-BO tuning of the FIS. This multi-fidelity modeling approach effectively balances theoretical rigor with engineering real-time feasibility.

1) KINEMATIC MODEL

The kinematic model neglects the vehicle's force dynamics and considers only geometric relationships of position, speed,

and heading angle. The mathematical formulation is:

$$\dot{X} = v_x \cos(\psi) \quad (1)$$

$$\dot{Y} = v_x \sin(\psi) \quad (2)$$

$$\dot{\psi} = \frac{v_x \tan(\delta)}{L} \quad (3)$$

where X, Y are the global coordinates, v_x is the longitudinal speed, ψ is the heading angle, δ is the steering angle, and L is the wheelbase. Although the model is computationally efficient, its accuracy degrades under high-speed or sharp-turn conditions.

2) LINEAR DYNAMIC MODEL

To achieve higher fidelity, the linear dynamic model explicitly incorporates lateral tire forces and yaw moment dynamics, making it suitable for moderate slip and steering regimes. The key equations are:

$$m\dot{v}_y = -mv_x\dot{\psi} + 2C_{af}\left(\delta - \frac{v_y + l_f\dot{\psi}}{v_x}\right) + 2C_{ar}\left(\frac{-v_y + l_r\dot{\psi}}{v_x}\right) \quad (4)$$

$$I_z\ddot{\psi} = 2l_f C_{af}\left(\delta - \frac{v_y + l_f\dot{\psi}}{v_x}\right) - 2l_r C_{ar}\left(\frac{-v_y + l_r\dot{\psi}}{v_x}\right) \quad (5)$$

where m is the vehicle mass, I_z is the yaw moment of inertia, v_x is the longitudinal velocity, v_y is the lateral velocity, ψ is the yaw angle, δ is the steering angle, l_f, l_r are the distances from the center of gravity to the front and rear axles, and C_{af}, C_{ar} are the cornering stiffnesses.

3) NONLINEAR DYNAMIC MODEL

To accurately capture vehicle behavior under large slip angles and high-speed conditions, a nonlinear dynamic model is adopted to account for the complex tire-road interactions. The vehicle dynamics are described using a three-degrees-of-freedom (3-DOF) structure, incorporating longitudinal, lateral, and yaw motions. As illustrated in Figure 1, the coordinate system, front and rear lateral tire forces, and slip angles provide the geometric foundation for the formulation of the nonlinear equations.

The general form of the discrete-time nonlinear system is given by

$$x_{k+1} = f(x_k, u_k, w_k), \quad y_k = h(x_k, v_k) \quad (6)$$

where $x_k \in \mathbb{R}^n$, $u_k \in \mathbb{R}^m$, and $y_k \in \mathbb{R}^p$ denote the state vector, control input, and system output at time step k , respectively. The terms w_k and v_k represent process disturbances and measurement noise, reflecting real-world uncertainties.

The continuous-time dynamic model incorporates the Brush tire model to characterize lateral tire forces under large slip conditions. The equations of motion are expressed

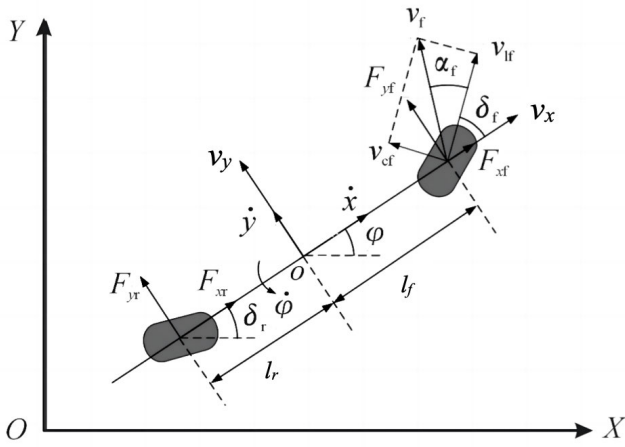


FIGURE 1. Three-degree-of-freedom vehicle dynamics model with front and rear lateral tire forces, yaw motion, and slip angles.

as follows

$$\begin{aligned} m\dot{v}_y &= -mv_x\dot{\psi} + F_{yf}\cos(\delta) + F_{yr} \\ m\dot{v}_x &= mv_y\dot{\psi} + F_{xf} - F_{yf}\sin(\delta) \end{aligned} \quad (7)$$

$$\begin{aligned} I_z\ddot{\psi} &= l_f F_{yf}\cos(\delta) - l_r F_{yr} \\ \dot{X} &= v_x\cos(\psi) - v_y\sin(\psi) \\ \dot{Y} &= v_x\sin(\psi) + v_y\cos(\psi) \end{aligned} \quad (8)$$

where F_{yf} and F_{yr} denote the front and rear lateral tire forces, respectively, and F_{xf} represents the front longitudinal tire force.

While a simple linear tire model is computationally efficient, it is only valid under small slip conditions and fundamentally fails to capture tire force saturation. Because automatic obstacle avoidance often involves aggressive emergency maneuvers that push the vehicle to its handling limits, a linear model would severely overestimate the available lateral grip, potentially leading to vehicle instability and control failure. Therefore, to ensure realistic predictions and safe control constraints under extreme dynamics, the nonlinear Brush tire model is adopted. The lateral tire force F_y is described by

$$F_y = \begin{cases} \mu F_z [3\theta\sigma - 3(\theta\sigma)^2 + (\theta\sigma)^3], & \text{if } \sigma \leq \sigma_s \\ \mu F_z, & \text{otherwise} \end{cases} \quad (9)$$

where μ is the road friction coefficient, F_z is the vertical load, θ represents the tire shape factor (a key parameter characterizing the comprehensive stiffness of the tire contact patch), σ is the equivalent slip angle, and σ_s is the saturation threshold. This nonlinear representation ensures accurate prediction of vehicle dynamics during aggressive maneuvers, which is essential for robust control in autonomous driving applications.

However, to maintain computational tractability and strict real-time feasibility for the MPC solver, the explicit friction circle constraint—where lateral tire force capacity intrinsically decreases during simultaneous hard braking or acceleration (i.e., combined slip)—is simplified in this model.

Consequently, while the adaptive fuzzy logic effectively adjusts the optimal control weights based on the nominal road friction indicator μ , it does not dynamically account for the instantaneous friction circle saturation. Addressing this tightly coupled dynamic constraint without compromising the QP solver's real-time efficiency remains a critical area for future refinement, which will further elevate the simulation fidelity for extreme emergency avoidance on low-adhesion surfaces.

To further enhance model fidelity, wheel rotational dynamics are incorporated to represent the generation of longitudinal forces during traction and braking. The angular velocity $\omega_{\diamond\star}$ of each wheel evolves as

$$\dot{\omega}_{\diamond\star} = \frac{1}{I^w}(T_{\diamond\star}^a + T_{\diamond\star}^b - R_e F_{\diamond\star}^l), \quad \forall \diamond\star \in \mathcal{F} \times \mathcal{S} \quad (10)$$

where I^w is the wheel moment of inertia, T^a and T^b are the applied acceleration and braking torques, F^l is the longitudinal tire force, and R_e is the effective rolling radius.

In a rear-wheel-drive configuration, the front wheels receive no drive torque:

$$T_{f\star}^a = 0, \quad \forall \star \in \mathcal{S} \quad (11)$$

while the total torque T^a is distributed to the rear wheels based on their angular velocity ratio:

$$T_{r\star}^a = T^a \left(1 - \frac{\omega_{r\star}}{\omega_{rl} + \omega_{rr}} \right), \quad \forall \star \in \mathcal{S} \quad (12)$$

Although the inclusion of wheel dynamics increases system stiffness and computational complexity, it significantly improves predictive accuracy under transient driving conditions, making it indispensable for high-fidelity simulation and control design in AVs.

B. FUNDAMENTALS OF MODEL PREDICTIVE CONTROL (MPC)

MPC is a state-of-the-art, optimization-based control strategy that leverages a prediction model of system dynamics and receding horizon optimization to generate control actions [23]. Owing to its predictive capability and explicit constraint handling, MPC has gained widespread adoption in autonomous vehicle systems for tasks such as trajectory tracking, collision avoidance, and real-time motion planning.

1) BASIC PRINCIPLE OF MPC

At each sampling instant, MPC predicts the system evolution over a finite prediction horizon N_p and computes an optimal sequence of control actions $U = \{u(0), u(1), \dots, u(N_c - 1)\}$, where $N_c \leq N_p$ is the control horizon. Only the first control input is applied before the process is repeated at the next time step, thus implementing a receding horizon policy. This feedback mechanism enables MPC to react effectively to disturbances, model uncertainties, and environmental changes.

The MPC optimization problem is formulated to minimize a multi-objective cost function composed of trajectory

tracking error, control effort, input rate variation, and a terminal penalty that ensures stability and robustness:

$$J(x_k, U) = \sum_{i=0}^{N_p-1} (\|x_{k+i|k} - x_{\text{ref},k+i}\|_Q^2 + \|u_{k+i|k} - u_{\text{ref},k+i}\|_R^2 + \|\Delta u_{k+i|k}\|_S^2) + V_f(x_{k+N_p|k}) \quad (13)$$

where Q , R , and S are weighting matrices that balance tracking performance, control effort, and smoothness, respectively. Specifically, the third term in the cost function, weighted by S , penalizes the control input increment $\Delta u_{k+i|k}$ (i.e., the rate of change of the steering command). This penalty is crucial for ensuring the smoothness of evasive maneuvers, thereby enhancing passenger comfort, preserving vehicle stability, and preventing severe actuator wear caused by high-frequency steering oscillations. Furthermore, the reference signals $x_{\text{ref},k+i}$ and $u_{\text{ref},k+i}$, along with the incremental input $\Delta u_{k+i|k}$, capture tracking and actuator dynamics [24]. The terminal cost $V_f(x_{k+N_p|k})$ guarantees long-term performance and closed-loop stability. MPC systematically enforces a set of operational and physical constraints, including:

$$\begin{aligned} x_{\min} \leq x_k \leq x_{\max}, \quad u_{\min} \leq u_k \leq u_{\max}, \\ \Delta u_{\min} \leq \Delta u_k \leq \Delta u_{\max}, \quad g(x_k, u_k) \leq 0 \end{aligned} \quad (14)$$

For real-time implementation in AVs, the continuous-time nonlinear system is typically linearized and discretized about an operating point, leading to the state-space form:

$$x(k+1) = Ax(k) + Bu(k) \quad (15)$$

where A and B represent the discrete-time system matrices.

Recent advances in real-time numerical solvers and onboard computational resources have further enhanced the practical feasibility of MPC in complex, high-dimensional, and fast-changing driving scenarios. The integration of learning-based methods—such as online model adaptation and adaptive parameter tuning—enables MPC to better handle time-varying uncertainties and disturbances.

Compared with traditional control methods (e.g., PID, LQR), MPC's predictive optimization and explicit constraint management provide essential advantages for safe and high-performance operation in automated driving. Nevertheless, challenges remain in the form of online computational burden, model uncertainty, and adaptation to highly variable road and traffic conditions.

To address these limitations, this paper introduces an adaptive fuzzy MPC framework in which the weights of the MPC cost function are dynamically adjusted by a FIS optimized via a dual-stage approach. This enables robust and high-precision control across diverse environments and driving scenarios, forming the theoretical basis for the subsequent methodology.

2) OBSTACLE MODELING AND PENALTY FUNCTION DESIGN
Robust obstacle modeling forms a critical component of autonomous vehicle control frameworks. In the broader field

of autonomous driving, state-of-the-art perception modules frequently employ LIDAR, camera, or radar systems to detect and localize obstacles [25]. Typically, clustering algorithms are applied to aggregate spatially close point cloud returns into coherent obstacle objects, allowing for effective differentiation between large, dense obstacles and scattered, isolated hazards [26].

However, the primary focus of this research is not on the perception or sensing layer, but on the design and evaluation of a MPC framework capable of robust obstacle avoidance given known obstacle positions and sizes [27]. In this context, the study assumes that reliable obstacle information is already available to the controller so that all efforts can be devoted to control-oriented modeling and penalty function optimization. To provide a more complete understanding of the method's applicability and limitations, it is essential to consider how dynamic obstacles might theoretically affect the proposed framework. While the current evaluations primarily rely on static obstacle setups to validate the control and optimization scheme, dynamic obstacles introduce complex, time-varying spatial constraints. Theoretically, safely navigating around moving obstacles would require integrating a robust trajectory prediction module directly into the MPC prediction horizon. This integration would not only increase the overall computational burden but also necessitate more rapid adaptations of the fuzzy weights to continuously accommodate sudden environmental changes and preserve closed-loop system stability.

To encode obstacle avoidance behavior within the MPC cost, a spatial penalty function is introduced. This function dynamically increases the avoidance cost as the vehicle approaches an obstacle, effectively guiding the control system to generate safe and smooth evasive maneuvers. A widely adopted form for the penalty function is:

$$J_{\text{obs},j} = \frac{S_{\text{obs}} v_i}{(x_i - x_0)^2 + (y_i - y_0)^2 + \zeta} \quad (16)$$

where S_{obs} is the penalty weight for the j -th obstacle, $v_i = v_x^2 + v_y^2$ denotes the vehicle's relative velocity, (x_i, y_i) are the coordinates of the obstacle point, (x_0, y_0) is the vehicle's center of mass, and ζ is a small positive constant for numerical stability.

A key aspect of this design is the selection of minimum distance thresholds and penalty scaling parameters. If the minimum allowable distance $d_{\min}$ is set too small, the cost function may become excessively sensitive, leading to over-reactive avoidance and degraded comfort. If too large, the system may fail to react to hazards in time, undermining safety.

In this research, the obstacle penalty parameters are systematically tuned and validated through comprehensive simulation experiments in Matlab-Simulink and CarSim. Various penalty landscape formulations are assessed, and the proposed third formulation demonstrates superior trajectory smoothness and safety under complex multi-obstacle scenarios. Adjustments are made iteratively based on closed-loop simulation feedback to ensure that the resulting control

strategy balances timely obstacle avoidance with vehicle stability and passenger comfort.

Overall, the proposed obstacle modeling strategy—grounded in proven penalty functions and informed by simulation-based tuning—provides a generic and perception-agnostic interface for integrating obstacle constraints into adaptive fuzzy MPC. This allows the control framework to maintain high tracking accuracy and operational safety, independent of the underlying sensing modality, making it broadly applicable to a range of autonomous vehicle architectures.

III. PROPOSED ADAPTIVE FUZZY MPC FRAMEWORK

A. FRAMEWORK OVERVIEW

Figure 2 illustrates the overall architecture of the proposed dual-stage adaptive fuzzy MPC framework for autonomous vehicle obstacle avoidance. The entire system is composed of an offline optimization phase and an online control phase. In the offline phase, PSO is first used to coarsely tune the membership function parameters of the FIS for key inputs such as road friction coefficient μ and lateral deviation e_y , followed by BO to fine-tune the output membership functions, specifically targeting the cost weights (ρ and Δu_{weight}).

The motivation for this dual-stage optimization strategy stems from the complementary strengths of PSO and BO in solving complex, non-convex tuning problems. The parameter space of a FIS is often high-dimensional and populated with numerous local optima. PSO, with its robust global exploration capabilities, effectively navigates this expansive landscape to identify the *basin of attraction* for the global optimum, thereby avoiding poor local minima. However, due to its stochastic nature, PSO is often computationally expensive and slow to converge when fine-tuning solutions within a specific region. This is where BO excels. By leveraging a probabilistic surrogate model (typically a Gaussian Process) and an acquisition function to guide the search, it minimizes the number of expensive objective function evaluations. This makes it highly sample-efficient for local refinement. Specifically, PSO is highly suited for the global coarse tuning of high-dimensional input MFs with a large search space, while BO enables the efficient local refinement of output MFs. This strategic assignment guarantees the high-precision MPC weight adjustment necessary to perfectly match both FIS and dynamic control requirements. Consequently, this hybrid approach effectively balances the broad exploration of PSO with the precise, low-cost exploitation of BO, yielding a more robust and optimal FIS tuning than either method could achieve in isolation.

This dual-stage approach produces an FIS that accurately captures the mapping from driving scenarios to optimal cost parameters. During online control, the trained FIS adaptively generates cost function weights in real time based on current obstacle information, road conditions, and updated vehicle state. These adaptive weights are directly integrated into the MPC controller, which incorporates modules for model

prediction, constraint encoding, cost function generation, and Quadratic Program (QP) solving. The controller outputs steering commands (δ , $\Delta\delta$) to update the vehicle trajectory. By systematically linking offline global optimization with real-time adaptive control, this framework achieves robust, constraint-aware obstacle avoidance and trajectory tracking across a wide range of challenging scenarios, while maintaining computational efficiency suitable for real-world autonomous driving applications.

The offline optimization stage employs PSO and Bayesian tuning to train the FIS membership functions, while the online control phase integrates obstacle and road condition information to adaptively adjust MPC cost weights, enabling real-time, constraint-aware vehicle control.

B. FUZZY-BASED ADAPTIVE WEIGHT MECHANISM

To enhance adaptability under varying speeds and road conditions, we integrate a fuzzy logic-based mechanism to adjust the MPC cost-function weights online. The implementation leverages the MATLAB Fuzzy Logic Toolbox for FIS design and tuning.

To ensure robust inference, five representative simulation scenarios were systematically defined by combining target vehicle speed and tire–road friction coefficient. These scenarios serve as a testbed for optimizing the membership functions of the FIS. A dual-stage optimization procedure was employed: PSO was first used for coarse tuning of the input membership functions, followed by BO to refine the output membership parameters.

The FIS uses the longitudinal velocity (V_x), lateral velocity (V_y), and road friction indicator (μ) as inputs and generates a set of weight scaling factors as outputs to adapt the MPC objective in real time:

$$Q_k = \alpha_Q(k) Q_0, \quad R_k = \alpha_R(k) R_0, \quad S_k = \alpha_S(k) S_0 \quad (17)$$

where Q_0 , R_0 , and S_0 denote baseline weight matrices, and $\alpha_Q(k)$, $\alpha_R(k)$, and $\alpha_S(k)$ are nonnegative scaling factors inferred by the FIS at time step k .

The fuzzy rules are constructed using Gaussian and triangular membership functions defined over the operating ranges of the input variables. Specifically, to define the structure of the fuzzy rule base, each input variable is partitioned into distinct linguistic terms (e.g., *Low*, *Medium*, and *High*). The comprehensive rule base is then formulated based on the combinations of these input states. The basis for this initial design is fundamentally grounded in expert engineering knowledge of vehicle handling dynamics prior to the optimization phase. For example, expert knowledge dictates that when the road friction μ is *Low* and the velocity V_x is *High*, the rule base initially assigns a *Large* weight to the steering rate penalty (α_S) to proactively prevent tire saturation and preserve lateral stability. A generic fuzzy rule within this specific structure can be expressed as

$$\text{IF } V_x \text{ is } A_i \text{ AND } V_y \text{ is } B_j \text{ AND } \mu \text{ is } D_\ell \text{ THEN } \alpha \text{ is } C_k \quad (18)$$

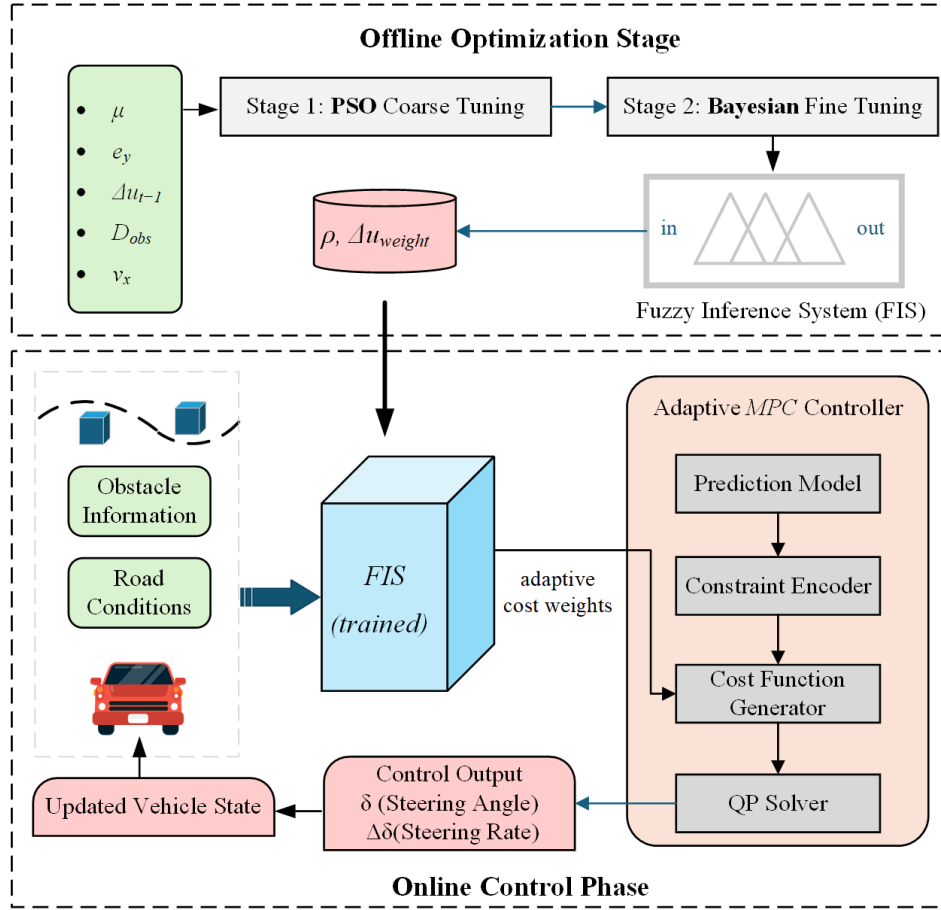


FIGURE 2. Overall framework of the proposed dual-stage optimized adaptive fuzzy MPC algorithm. The offline optimization stage employs PSO and Bayesian tuning to train the FIS membership functions, while the online control phase integrates obstacle and road condition information to adaptively adjust MPC cost weights, enabling real-time, constraint-aware vehicle control.

or mathematically denoted as

$$\mu_{A_i}(V_x), \mu_{B_j}(V_y), \mu_{D_l}(\mu) \rightarrow \mu_{C_k}(\alpha) \quad (19)$$

This fuzzy-based adaptation allows the controller to balance tracking accuracy, safety, and smoothness across diverse conditions without requiring explicit re-tuning of cost weights.

For fairness and real-time feasibility, the prediction and control horizons are kept fixed throughout all experiments, with $N_p = 50$ and $N_c = 15$.

$$\min_{\theta_{MF}} J_{\text{PSO}} = \sqrt{\frac{1}{N} \sum_{k=1}^N (e_{y,k}^2 + e_{\phi,k}^2)} \quad (20)$$

where θ_{MF} denotes the set of all input membership function parameters, $e_{y,k}$ and $e_{\phi,k}$ represent the lateral and heading errors at step k , and N is the total number of simulation steps. This formulation ensures that the tuned FIS achieves robust tracking performance across diverse operating conditions. Following this, BO is applied to fine-tune the membership functions of the fuzzy output variable (ρ), which modulates the cost function weights within the MPC framework.

The objective function remains consistent:

$$\min_{\theta_\rho} J_{\text{BO}} = \sqrt{\frac{1}{N} \sum_{k=1}^N (e_{y,k}^2 + e_{\phi,k}^2)} \quad (21)$$

Here, (θ_ρ) refers to the output membership function parameters. BO offers a probabilistic global search strategy using surrogate modeling (e.g., Gaussian processes) and acquisition functions to balance exploration and exploitation. It efficiently identifies promising parameter combinations in the high-dimensional fuzzy parameter space.

Figure 3 summarizes the BO results and the corresponding evolution of the output membership functions. The response surface over the selected design variables r_3 and r_4 (sampled evaluations marked by red dots and the best solution highlighted by a blue star) reveals a highly non-convex objective landscape. Here, r_3 and r_4 are design parameters rather than physical quantities; they determine the peak locations of the *Medium* and *Large* fuzzy sets for the output variable ρ , thereby shaping the controller's output response.

The optimized membership functions for ρ are compared across three stages: the initial design, the PSO-based coarse

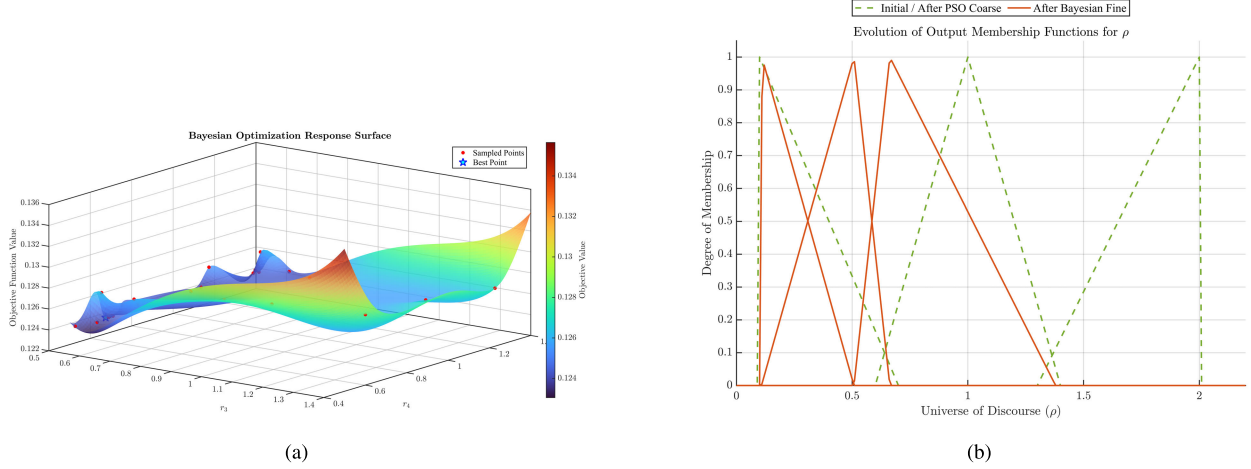


FIGURE 3. Two-stage fuzzy output optimization: (a) Bayesian response surface; (b) evolution of output membership functions.

tuning, and the final Bayesian refinement. The final configuration exhibits sharper transitions and improved separation among linguistic terms, which supports more responsive and accurate online weight adaptation.

After completing both stages, the tuned FIS is exported and embedded into the online MPC framework, where it performs real-time inference to adapt the MPC cost-function weights according to evolving vehicle states and environmental conditions. The overall two-stage tuning procedure (PSO coarse search followed by Bayesian refinement) is summarized in Algorithm 1.

Algorithm 1 Multi-Scenario Adaptive FIS Tuning via PSO and BO

Input: Initial FIS structure

Output: Robust FIS with optimized membership functions

- 1: Scenario set $\mathcal{S}$ (e.g., icy, wet, dry)
 - 2: **Step 1: PSO Coarse Tuning**
 - 3: Optimize input membership functions using PSO
 - 4: Objective: Minimize mean RMS error across all scenarios in $\mathcal{S}$
 - 5: **Step 2: Bayesian Fine Tuning**
 - 6: Optimize output membership functions using BO
 - 7: Objective: Further minimize mean RMS error across $\mathcal{S}$
 - 8: **Step 3: Multi-Scenario Validation**
 - 9: **for** each scenario in $\mathcal{S}$ **do**
 - 10: Simulate adaptive MPC with tuned FIS
 - 11: Record RMS tracking errors and assess stability
 - 12: **end for**
 - 13: **Step 4: Visualization**
 - 14: Plot optimization surfaces and MF evolution for analysis
-

C. ONLINE ADAPTIVE MPC IMPLEMENTATION

During online operation, the tuned FIS dynamically adjusts the key parameters of the MPC framework according to real-time measurements. These measurements include vehicle lateral deviation (e_y), estimated road friction coefficient (μ), the rate of change of the previous control input (Δu_{prev}), normalized obstacle distance ($d_{\text{obs,norm}}$), and normalized vehicle speed ($v_{x,\text{norm}}$). All input variables are normalized

to enhance the robustness of fuzzy inference across varying scenarios.

As summarized in Algorithm 2, the FIS produces two primary outputs at each control step: the base input weighting factor ρ_{base} and the control input variation weight $R_{\Delta u, \text{final}}$. These outputs are then adaptively refined using scenario-dependent scaling factors, such as $\rho_{\text{obs_factor}}$ (applied when obstacles are present) or $\rho_{\text{clear_factor}}$ (when the path is clear), as well as adjustments based on the current road friction and reference velocity. The final input weighting matrix is computed as $R_k = (\rho_{\text{base}} \cdot \rho_{\text{current_factor}})I$, where I is the identity matrix, and the input variation matrix is set as $R_{\Delta u} = R_{\Delta u, \text{final}}I$.

Algorithm 2 Adaptive MPC Batch Simulation and Controller Core

Input: Scenario set $\mathcal{S}$ (varied speeds/friction/obstacles), final FIS parameters

Output: Tracking and control performance metrics

- 1: Initialize results table
 - 2: **for** each scenario (v, μ) in $\mathcal{S}$ **do**
 - 3: Set up vehicle parameters, constraints, and initial state
 - 4: **for** each simulation step **do**
 - 5: **Fuzzy Inference:**
 - 6: Input current state and environment to FIS to get $\rho, R_{\Delta u}$
 - 7: **Adaptive MPC Core:**
 - 8: (1) Build and update prediction and cost matrices
 - 9: (2) If Discrete Algebraic Riccati Equation (DARE) fails, fallback to manual calculation for terminal cost/gain
 - 10: (3) For each horizon step, adapt R according to obstacle status
 - 11: (4) Formulate and solve constrained QP for optimal control
 - 12: (5) Update state, input, and log outputs
 - 13: **end for**
 - 14: Compute RMS metrics (lateral/heading error, control input, input rate)
 - 15: Store results for scenario
 - 16: **end for**
 - 17: Summarize and analyze all results
-

In addition, the adaptive safety margin s_m is dynamically updated according to the baseline margin, road friction, and vehicle speed, ensuring real-time constraint satisfaction for obstacle avoidance. This closed-loop adaptation of MPC parameters, driven by fuzzy inference, enables the controller to maintain high-precision trajectory tracking and robust obstacle avoidance under diverse and rapidly changing operating conditions.

The adaptive MPC controller employs a discretized state-space model with these parameters to compute optimal control actions at each time step. The state vector is defined as $\mathbf{x} = [v_y, r, \psi, e_y]^T$, where v_y is lateral velocity, r is yaw rate, ψ is yaw angle error, and e_y is lateral deviation. The continuous-time system matrices are given by

$$A_c = \begin{bmatrix} -\frac{C_{f,\alpha} + C_{r,\alpha}}{v_{x,ref} m} - v_{x,ref} + \frac{C_{f,\alpha} a - b C_{r,\alpha}}{v_{x,ref} m} & 0 & 0 \\ \frac{a C_{f,\alpha} - b C_{r,\alpha}}{I_z v_{x,ref}} - \frac{a^2 C_{f,\alpha} + b^2 C_{r,\alpha}}{I_z v_{x,ref}} & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & v_{x,ref} \end{bmatrix} \quad (22)$$

$$B_c = \begin{bmatrix} \frac{C_{f,\alpha}}{m} & 0 \\ \frac{a C_{f,\alpha}}{I_z} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (23)$$

where m is the vehicle mass, I_z is the yaw moment of inertia, a and b are the distances from the center of gravity to the front and rear axles, $v_{x,ref}$ is the reference longitudinal velocity, and $C_{f,\alpha}$, $C_{r,\alpha}$ are the front and rear tire cornering stiffnesses, which are adjusted online based on μ . The system is discretized using Zero-Order Hold (ZOH) with sampling time T_s to obtain matrices A and B . The cost function for the MPC is formulated as

$$J = \sum_{k=0}^{N_p-1} \left(\|x_{k|i} - x_{ref,k|i}\|_Q^2 + \|u_{k|i}\|_{R_k}^2 + \|\Delta u_{k|i}\|_{R_{\Delta u}}^2 \right) + \|x_{N_p|i} - x_{ref,N_p|i}\|_P^2 \quad (24)$$

where $x_{k|i}$ and $u_{k|i}$ are the predicted states and control inputs at step k (computed at time i), $x_{ref,k|i}$ is the reference trajectory (typically zero for deviation states). The weighting matrices are:

- $Q = C^T C$, where C is the output selection matrix (e.g., selecting e_y).
- $R_k = (\rho_{base} \cdot \rho_{current_factor}) I$.
- $R_{\Delta u} = R_{\Delta u, final} I$.
- P is the terminal weight, computed via the DARE to ensure stability.

The discrete-time system is:

$$x(k+1|i) = Ax(k|i) + Bu(k|i) \quad (25)$$

Constraints are updated in real time according to the current road and obstacle scenario. State constraints ($x_{min} \leq x(k|i) \leq x_{max}$) and input constraints ($u_{min} \leq u(k|i) \leq u_{max}$) are enforced, with time-invariant constraints

(e.g., lane boundaries) and time-varying obstacle constraints generated dynamically. The obstacle constraints take the form $f_{x,Tv}x(k|i) \leq v_{x,Tv}$, e.g., for a right-side obstacle, $e_y \leq y_{obs,min} - s_m$.

At each sampling instant i , the controller solves a QP to obtain the optimal input sequence $U_i^* = [u_{0|i}^*, u_{1|i}^*, \dots, u_{N_p-1|i}^*]$ over the prediction horizon N_p . Only the first input $u_{0|i}^*$ is applied, and the process repeats at the next time step (receding horizon). If the QP solver fails, the previous control input is retained to ensure safe fallback.

IV. RESULTS AND EVALUATION

This section evaluates the proposed controller through comprehensive simulations and compares it with the baseline MPC and a PID controller. We first describe the simulation setup and statistical protocol (10 runs per scenario, reported as mean $\pm$ std), and then report quantitative tracking and control-effort results together with real-time runtime analysis.

A. SIMULATION SETUP

All simulations were conducted in MATLAB R2023b on a workstation with Intel Core i9-12900K CPU and 32 GB RAM to ensure reproducibility, utilizing a nonlinear 3-DOF vehicle dynamics model to capture realistic handling limits. The proposed adaptive fuzzy MPC algorithm, along with baseline MPC and PID controllers, was independently implemented and evaluated across five representative scenarios: three low-speed cases ($v_x = 8 \text{ m/s}$) on icy, wet, and dry roads ($\mu = 0.30, 0.50, 0.85$), and two higher-speed cases ($v_x = 13.9 \text{ m/s}, 22 \text{ m/s}$) on dry surfaces. In all trials, three static obstacles were introduced at fixed time intervals to challenge each controller. A small initial lateral offset ($e_y(0) = 0.2 \text{ m}$) was applied to excite both lateral and heading dynamics, testing each controller's disturbance rejection and recovery. All environmental parameters and obstacle configurations were strictly standardized across all tests to guarantee a fair and consistent benchmarking assessment.

The core vehicle model, whose physical parameters are listed in Table 1, remained constant; tire stiffness was updated online according to friction in each scenario. All experiments used a 0.1 s sampling interval, a 50-step prediction horizon, and a 20 s simulation duration, with a lane width of 3 m . To reflect practical variations and statistical reliability, each scenario was repeated 10 times with randomized initial headings and obstacle timings. Unless otherwise stated, all reported performance metrics are presented as mean $\pm$ standard deviation (std) over the 10 runs.

For fairness, all controllers shared the same plant model, initial conditions, constraints, and obstacle settings. The adaptive fuzzy MPC incorporated a real-time FIS trained offline via two-stage PSO and BO for dynamic cost adjustment, while the baseline MPC and PID employed fixed parameters. To ensure the strict reproducibility of this offline training phase, the specific technical details

and hyperparameter recommendations for the PSO and BO algorithms are summarized in Table 2.

For the PSO stage, a swarm size of 30 and a maximum of 40 iterations are recommended. This configuration provides a sufficient computational budget to robustly explore the high-dimensional FIS parameter space and identify the basin of attraction without incurring excessive computational overhead. Furthermore, an adaptive inertia weight w that linearly decreases from 0.9 to 0.4 is implemented to facilitate aggressive global exploration initially, while encouraging precise local exploitation in later stages. The cognitive and social acceleration coefficients (c_1 and c_2) are symmetrically set to 2.0. In the subsequent BO fine-tuning stage, a Gaussian Process (GP) with a Matérn 5/2 kernel is employed as the surrogate model, as it relaxes the assumption of infinite differentiability and excels at modeling physical dynamics with abrupt gradient changes. Crucially, the Expected Improvement Plus (EI Plus) is utilized as the acquisition function to actively force exploration when the search is trapped in flat local regions. A maximum of 25 BO evaluations is strictly sufficient to achieve high-precision local refinement, given the excellent starting point established by PSO.

TABLE 1. Physical parameters of the vehicle model used in simulation.

Parameter	Symbol	Value	Unit
Vehicle mass	m	2050	kg
CG to front axle distance	l_f	1.2	m
CG to rear axle distance	l_r	1.5	m
Yaw moment of inertia	I_z	3344	kg·m ²
Front cornering stiffness (per tire)	C_{af}	4.5×10^4	N/rad
Rear cornering stiffness (per tire)	C_{ar}	4.0×10^4	N/rad
Gravity	g	9.81	m/s ²
Road friction coefficient	μ	0.30 / 0.50 / 0.85	–
Sampling time	T_s	0.1	s
Prediction horizon	N_p	50	–
Lane width	W_{lane}	3.0	m

TABLE 2. Hyperparameter Settings and Recommendations for the PSO-BO Algorithm.

Algorithm	Parameter	Value / Selection
PSO	Swarm Size (N_{pop})	30
	Maximum Iterations ($Iter_{\text{max}}$)	40
	Inertia Weight (w)	Linearly decreasing (0.9 to 0.4)
	Acceleration Coefficients (c_1, c_2)	2.0, 2.0
BO	Surrogate Model	GP
	Kernel Function	Matérn 5/2
	Acquisition Function	Expected Improvement Plus
	Maximum Evaluations	25

The controller mode-switching logic was consistent throughout, and the impact of initial state perturbation was analyzed to evaluate robustness and recovery capability.

Upon completion of all batch simulations, results were aggregated for comprehensive quantitative and visual comparison, as reported in the following section.

B. COMPUTATIONAL COST AND REAL-TIME FEASIBILITY

To assess real-time feasibility under $T_s = 0.1$ s, we recorded the total wall-clock execution time t_{solve} at each control step—which explicitly encompasses both the fuzzy logic inference duration and the MPC QP solution time—for the proposed controller and the baseline MPC. All timing measurements were obtained on the same workstation as in the simulation setup and exclude simulation rendering and data logging overhead. The underlying QP was solved using MATLAB's `quadprog` (interior-point-convex) with warm-start enabled by shifting the previous optimal control sequence.

Across all scenarios and repetitions, a total of $N = 10,000$ control steps were collected (20 s per run, $T_s = 0.1$ s, 5 scenarios $\times$ 10 repetitions). The proposed controller achieved a mean / P95 / maximum t_{solve} of 20.6 ms / 45.3 ms / 83.7 ms, respectively. The deadline miss rate, defined as $\Pr(t_{\text{solve}} > T_s)$, was 0% over all recorded steps. For the baseline MPC, the mean / P95 / maximum t_{solve} was 21.4 ms / 47.8 ms / 88.9 ms, also with 0% deadline misses.

To reduce computational burden, we adopted (i) warm-start initialization, and (ii) a shorter control horizon $N_c = 15$ ($N_c \leq N_p$) while keeping the prediction horizon fixed at $N_p = 50$. Overall, the measured runtime statistics confirm that the proposed controller can be executed reliably in real time with $T_s = 0.1$ s and $N_p = 50$ across all tested scenarios.

While the aforementioned runtime statistics demonstrate robust real-time feasibility on a high-performance workstation, deploying the algorithm to an actual vehicle controller (e.g., an automotive Electronic Control Unit (ECU) or a rapid prototyping target like dSPACE) introduces practical hardware constraints. Actual vehicle controllers typically possess significantly lower raw computational power than desktop processors. However, the proposed AFMPC architecture is structurally highly amenable to embedded deployment. First, the most computationally intensive optimization burden (PSO-BO) is completely decoupled and handled offline, leaving only the lightweight FIS evaluation and a convex QP to be solved online. Second, while the simulation relies on MATLAB's general-purpose `quadprog` within an interpreted environment, industrial deployment involves generating highly optimized, compiled C/C++ code and utilizing specialized, embedded-grade QP solvers (e.g., OSQP or qpOASES). The massive efficiency gains from compiled code and embedded solvers effectively offset the reduced clock speeds of onboard hardware, ensuring the computation time remains strictly bounded within the $T_s = 0.1$ s sampling interval. Nevertheless, comprehensively validating the sim-to-real performance via Hardware-in-the-Loop (HIL) testing and full-scale vehicle implementation constitutes our immediate future work.

C. PERFORMANCE METRICS

The effectiveness of each control strategy is assessed using a comprehensive set of quantitative indicators, reflecting trajectory tracking precision, control effort, and actuation smoothness throughout the full simulation horizon. The primary metrics include the root mean square (RMS) of the lateral error (e_y), which measures the vehicle's average deviation from the lane centerline, and the RMS of the heading error (e_ϕ), characterizing the ability to maintain the desired yaw angle relative to the reference trajectory. Additionally, the RMS value of the steering angle command (δ) is computed to evaluate overall control activity and aggressiveness, while the RMS of the steering rate ($\Delta\delta$) serves as an indicator of actuation smoothness and passenger comfort. These metrics are calculated as follows:

$$\text{RMS}_{e_y} = \sqrt{\frac{1}{N} \sum_{k=1}^N e_{y,k}^2}, \quad \text{RMS}_{e_\phi} = \sqrt{\frac{1}{N} \sum_{k=1}^N e_{\phi,k}^2} \quad (26)$$

$$\text{RMS}_\delta = \sqrt{\frac{1}{N} \sum_{k=1}^N \delta_k^2}, \quad \text{RMS}_{\Delta\delta} = \sqrt{\frac{1}{N-1} \sum_{k=2}^N (\delta_k - \delta_{k-1})^2} \quad (27)$$

Lower values of lateral and heading error RMS correspond to higher trajectory tracking accuracy, while reduced steering input and steering rate RMS indicate more efficient and smoother actuation. This multi-dimensional evaluation provides a systematic basis for comparing the adaptive fuzzy MPC with baseline control schemes under diverse and challenging operating conditions.

D. EXPERIMENTAL RESULTS AND COMPARATIVE ANALYSIS

Figure 4 compares the closed-loop responses of the proposed adaptive fuzzy MPC with a baseline MPC and a conventional PID controller under five obstacle-rich scenarios spanning high-speed dry asphalt ($v_x=22$ m/s, $\mu=0.85$) to low-adhesion wet ($\mu=0.50$) and icy ($\mu=0.30$) roads. Each subfigure reports the lateral tracking error $e_y(t)$, heading error $e_\phi(t)$, and steering command $\delta(t)$ (denoted as u_1), while the shaded regions indicate time intervals in which obstacle-avoidance constraints are active. Notably, an unstable trend is observed for the PID controller in these scenarios; its exceptionally high tracking errors stem from system instability under varying vehicle speeds and road friction coefficients. This is primarily due to the PID controller's inherent lack of predictive capacity for nonlinear vehicle dynamics, as well as its inability to robustly adapt to the spatial and dynamic constraints imposed by abrupt obstacle avoidance maneuvers.

In addition to the three primary baselines, two references are included for context: a nominal-point LQR as a linear benchmark and an NMPC benchmark that optimizes the nonlinear vehicle model, providing a high-fidelity reference at the expense of higher computational cost.

Across all scenarios, the adaptive fuzzy MPC achieves the most favorable trade-off between tracking accuracy and control smoothness, with smaller peak and residual errors in both $e_y(t)$ and $e_\phi(t)$ and reduced oscillations in $\delta(t)$ during and after obstacle engagement. The advantage becomes more evident under low- μ conditions, where online fuzzy inference adjusts the MPC weights (e.g., robustness-related ρ and steering-rate penalty) to avoid overly aggressive steering actions that may trigger additional tire slip. In contrast, the baseline MPC uses fixed weights and tends to exhibit larger deviations and/or more pronounced steering fluctuations when operating closer to handling limits. The PID controller shows the weakest robustness, with large tracking errors and degraded stability once the system departs from the regime where linear error feedback remains effective.

Figure 5 summarizes the RMS safety and control-effort metrics for all controllers across five scenarios. Overall, the proposed Adaptive Fuzzy MPC achieves the best *overall* performance, delivering the smallest tracking errors (RMS of e_y and e_ϕ) while maintaining low control effort (RMS of u_1 and Δu_1), thereby providing a superior accuracy-smoothness trade-off compared with the Baseline MPC, PID, LQR, and NMPC benchmarks.

Specifically, the PID controller exhibits tracking errors that are orders of magnitude larger than all model-based methods, indicating a lack of robustness to obstacle-induced reference changes and nonlinear/constraint-dominated dynamics. LQR, while computationally lightweight, does not explicitly handle constraints and obstacle-driven objectives and thus cannot maintain competitive safety performance under the tested conditions. NMPC is able to enforce nonlinear dynamics but typically incurs a higher computational burden; in our experiments it does not translate this complexity into superior closed-loop accuracy and smoothness relative to the proposed controller. In contrast, the Baseline MPC improves substantially over PID/LQR but is limited by fixed cost weights, which leads to more aggressive steering activity under challenging conditions. The advantage of the proposed method is most pronounced under low adhesion (e.g., the low-icy case), where online fuzzy reweighting increases the penalty on input-rate variations and strengthens robustness-related terms when reduced tire grip is detected, effectively mitigating oscillatory steering and reducing both u_1 and Δu_1 without sacrificing e_y and e_ϕ .

These results demonstrate that the proposed Adaptive Fuzzy MPC consistently outperforms all competing controllers in terms of safety and comfort metrics across a wide range of speeds and friction levels, supporting its practicality for real-time trajectory tracking and obstacle avoidance.

The quantitative performance comparison is summarized in Table 3. Inspecting the tracking metrics, the proposed Adaptive MPC demonstrates superior robustness against friction heterogeneity, particularly in challenging low-adhesion environments. It reduces the RMS lateral tracking error (e_y) by up to 37% compared to the Baseline MPC (e.g., in the Low-wet scenario), while consistently maintaining heading

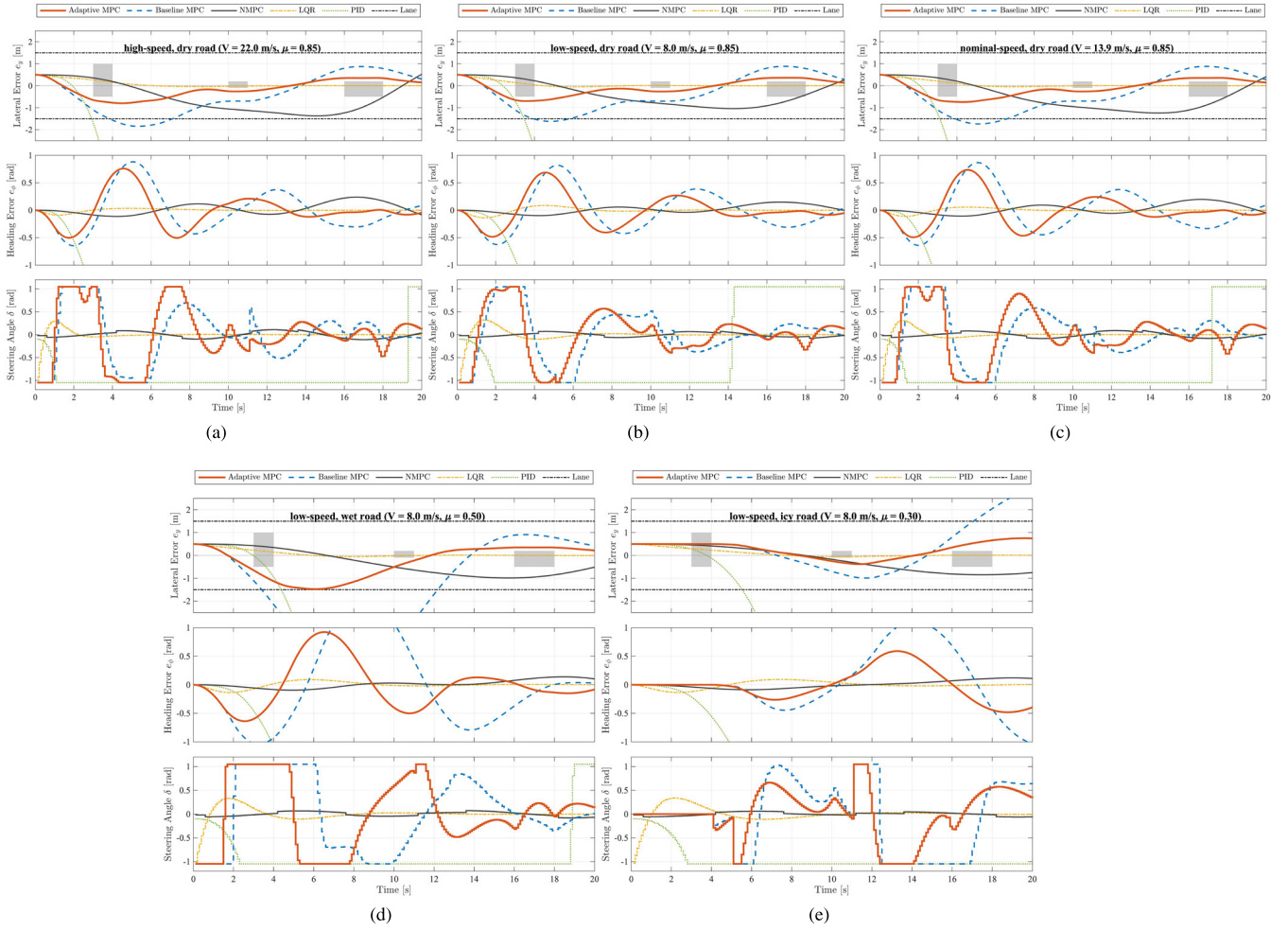


FIGURE 4. Performance comparison of Adaptive MPC, Baseline MPC, and PID controllers in five obstacle-rich scenarios.

TABLE 3. RMS performance (mean $\pm$ std over 10 runs) under five scenarios.

Scenario	Metric	Proposed	Base MPC	PID	LQR	NMPC
Low-icy ($v=8$ m/s, $\mu=0.30$)	e_y (m)	0.76 ± 0.05	0.80 ± 0.08	42.63 ± 12.47	0.92 ± 0.07	0.85 ± 0.06
	e_ϕ (rad)	0.29 ± 0.03	0.44 ± 0.06	5.85 ± 1.62	0.38 ± 0.04	0.32 ± 0.03
	u_1	0.58 ± 0.04	0.65 ± 0.07	0.91 ± 0.15	0.42 ± 0.05	0.36 ± 0.04
	Δu_1	0.71 ± 0.06	1.50 ± 0.18	0.26 ± 0.09	0.38 ± 0.05	0.51 ± 0.06
Low-wet ($v=8$ m/s, $\mu=0.50$)	e_y (m)	0.65 ± 0.04	1.03 ± 0.09	91.16 ± 22.35	0.82 ± 0.06	0.74 ± 0.05
	e_ϕ (rad)	0.18 ± 0.02	0.35 ± 0.05	7.45 ± 2.08	0.30 ± 0.03	0.25 ± 0.03
	u_1	0.42 ± 0.03	0.62 ± 0.06	0.92 ± 0.14	0.38 ± 0.04	0.32 ± 0.04
	Δu_1	0.68 ± 0.05	0.95 ± 0.12	0.28 ± 0.10	0.35 ± 0.04	0.48 ± 0.05
Low-dry ($v=8$ m/s, $\mu=0.85$)	e_y (m)	0.59 ± 0.03	0.70 ± 0.06	175.53 ± 38.62	0.69 ± 0.05	0.65 ± 0.04
	e_ϕ (rad)	0.15 ± 0.02	0.27 ± 0.04	8.53 ± 2.41	0.22 ± 0.03	0.19 ± 0.02
	u_1	0.26 ± 0.02	0.41 ± 0.05	0.93 ± 0.13	0.34 ± 0.04	0.28 ± 0.03
	Δu_1	0.43 ± 0.04	0.68 ± 0.09	1.05 ± 0.22	0.32 ± 0.04	0.42 ± 0.05
Nominal-dry ($v=13.9$ m/s, $\mu=0.85$)	e_y (m)	0.60 ± 0.04	0.74 ± 0.07	252.91 ± 51.84	0.75 ± 0.06	0.71 ± 0.05
	e_ϕ (rad)	0.16 ± 0.02	0.28 ± 0.04	10.75 ± 3.15	0.25 ± 0.03	0.21 ± 0.03
	u_1	0.28 ± 0.03	0.42 ± 0.05	0.94 ± 0.12	0.36 ± 0.04	0.30 ± 0.03
	Δu_1	0.52 ± 0.05	0.75 ± 0.10	1.51 ± 0.35	0.34 ± 0.04	0.45 ± 0.05
High-dry ($v=22$ m/s, $\mu=0.85$)	e_y (m)	0.60 ± 0.04	0.77 ± 0.08	306.24 ± 62.73	0.88 ± 0.07	0.82 ± 0.06
	e_ϕ (rad)	0.16 ± 0.02	0.28 ± 0.05	11.69 ± 3.48	0.29 ± 0.04	0.24 ± 0.03
	u_1	0.31 ± 0.03	0.45 ± 0.06	0.95 ± 0.14	0.39 ± 0.05	0.33 ± 0.04
	Δu_1	0.65 ± 0.06	0.92 ± 0.13	0.20 ± 0.08	0.37 ± 0.05	0.49 ± 0.06

Note: All controllers share the same steering limit ($|\delta| \leq 0.95$ rad $\approx 54^\circ$). PID often saturates in aggressive scenarios, causing lower variation in u_1 .

deviations (e_ϕ) below 0.3 rad. Note that both MPC variants outperform the conventional PID controller by an order of

magnitude, highlighting the necessity of predictive control in limit handling.

Crucially, the improved tracking accuracy of the proposed method does not come at the cost of aggressive actuation. Instead, the adaptive scheme exhibits higher control efficiency, lowering the RMS steering command (u_1) by approximately 11–37% across all test cases. More importantly, the steering rate (Δu_1) is significantly curtailed—for instance, dropping from 1.50 to 0.71 in the Low-icy scenario. This indicates that the online cost reweighting mechanism effectively suppresses the high-frequency control oscillations observed in the fixed-weight baseline, thereby achieving a superior balance between tracking precision and actuation smoothness.

E. COMPUTATIONAL EFFICIENCY AND REAL-TIME FEASIBILITY

In addition to control performance, real-time feasibility is essential for onboard deployment. We therefore measured the per-step wall-clock computation time for all controllers under the same simulation platform and sampling period $T_s = 0.1$ s. Results are aggregated over five scenarios with

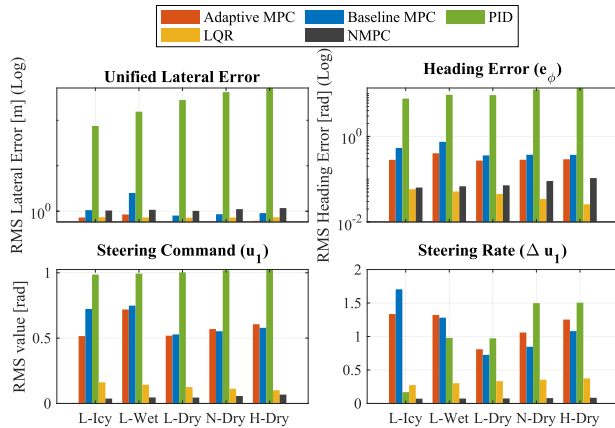


FIGURE 5. RMS comparison of Adaptive MPC, Baseline MPC, and PID controllers for five driving scenarios.

10 runs per scenario and exclude visualization and data-logging overhead.

Fig. 6 compares the average computation time on a logarithmic scale. PID and LQR exhibit sub-millisecond runtime, as they involve no online optimization. In contrast, MPC-based controllers solve a QP at each step. AFMPC and the baseline MPC show comparable runtime (tens-of-milliseconds level), indicating that the additional FIS evaluation introduces only marginal overhead relative to the QP solve.

NMPC is substantially more expensive, operating with a much smaller time margin to the real-time budget. Although the mean runtime may remain below T_s in the tested scenarios, strict real-time validation requires tail-latency reporting. Accordingly, we additionally report percentile statistics (P95/max) and the deadline-miss rate $\Pr(t_{\text{solve}} > T_s)$ to characterize worst-case behavior and real-time reliability.

Overall, AFMPC achieves a favorable accuracy–efficiency trade-off: it improves tracking accuracy and control

smoothness over the baseline MPC while maintaining a comfortable real-time margin at $T_s = 0.1$ s.

V. CONCLUSION

This paper proposed a AFMPC framework for autonomous vehicle obstacle avoidance and trajectory tracking. A FIS was tuned offline via PSO followed by BO, and was then deployed online to adapt the MPC objective in real time by reweighting tracking, robustness, and smoothness terms according to vehicle states and road adhesion conditions. This design explicitly addresses the trade-off between tracking accuracy and stability/comfort when operating near handling limits.

Comprehensive simulations over five representative scenarios demonstrated that the proposed AFMPC provides the best overall safety–comfort trade-off among all compared controllers. In particular, AFMPC achieved the lowest RMS lateral and heading errors while simultaneously reducing steering effort and steering-rate activity, with the advantage being most evident under low-adhesion conditions where fixed-weight MPC tends to induce more aggressive steering corrections. Compared with Baseline MPC, AFMPC reduced the RMS lateral error by 6–38% and improved heading accuracy by up to 49%, while decreasing steering effort by 12–39%. Moreover, relative to PID, LQR, and NMPC benchmarks, AFMPC maintained superior tracking performance with smoother control actions, indicating improved robustness to obstacle-induced reference changes and nonlinear/constraint-dominated dynamics. From an implementation perspective, the runtime statistics confirmed stable real-time operation at $T_s = 0.1$ s with $N_p = 50$, supporting the practical feasibility of AFMPC for onboard deployment.

It should be noted that because the intensive offline PSO-BO tuning relies on the specific dynamics of the vehicle, the proposed approach is inherently vehicle-dependent. While the robust predictive nature of the MPC can accommodate minor parametric variations, switching to a fundamentally different vehicle platform or tire model would require re-conducting the offline tuning process to ensure optimal performance. Consequently, future work will focus on the following directions: (i) validation on a real vehicle platform with Hardware-in-the-Loop (HIL) and closed-loop experiments, (ii) extension to dynamic obstacles under perception uncertainty and time-varying constraints, (iii) tighter integration with perception and state-estimation modules to improve robustness under sensing noise and unmodeled disturbances, and (iv) the development of an automated offline calibration framework to rapidly adapt the FIS parameters when migrating the algorithm to different vehicle architectures.

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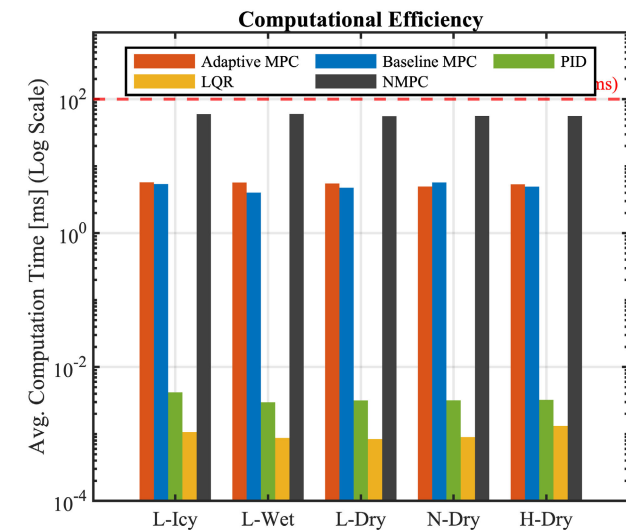


FIGURE 6. Average per-step computation time (log scale) of five controllers across five scenarios. The red dashed line denotes the real-time budget ($T_s = 100$ ms).

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